

TABLE I
FOUR CATEGORIES DEFINED BY INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

	CAT IV	CAT III	CAT II	CAT I
Description	For measurements performed on all sources of low-voltage installation	For measurements performed while building installation	For measurements performed on circuits directly connected to low-voltage installation	For measurements performed on circuits not directly connected to mains
Examples	Overhead wires and cable systems	Distribution boards and circuit breakers	Appliances such as portable equipment	Signal-level circuits of electronic PCBs
Rated voltage	Transient voltage			
150V	4000V	2500V	1500V	800V
300V	6000V	4000V	2500V	1500V
600V	8000V	6000V	4000V	2500V
1000V	12000V	8000V	6000V	4000V

sider looking for a DMM that offers *avant-garde* capabilities to befit your application needs. These *avant-garde* DMM functions save time when performing troubleshooting tasks. Here are some of the advanced capabilities designed to soothe the job of engineers and technicians.

Vsense. Voltage sense detector (Vsense) is a non-contact voltage detector in insulated wires, wall receptacles, fuses, junction boxes, switches and more that detects the presence of AC voltages nearby. If the presence of AC voltage is sensed, the multimeter's beeper sounds and the LED turns on.

Smart ohm. This (offset compensation) is another feature designed to remove unexpected DC voltages within the instrument at the input or at the circuit being measured, where resistance measurement errors are introduced.

Data logging. This feature makes it easy for factory maintenance personnel to generate a range of troubleshooting, monitoring and process documentation. The personnel can also collect data over an extended duration to support installation or repair of factory systems such as heating, ventilation and air-conditioning.

LPF. A regular true-rms DMM cannot measure output from a motor drive because the variable frequency drive applies pulse-width-modulated,

non-sinusoidal voltage to the motor terminals. The low-pass filter is designed to help block unwanted voltages above 1kHz when measuring AC voltage or AC frequency.

Harmonic ratio. This feature can help technicians quickly check for the presence of harmonics in the electrical power system. A pure sinusoidal waveform without harmonics has a harmonic ratio of zero per cent.

Z_{LOW} (low impedance mode). With dual-impedance meters, technicians can safely troubleshoot sensitive electronic or control circuits as well as circuits that may contain ghost voltages. They can more reliably determine whether voltage is present on a circuit.

With low-impedance input to the circuit under test reduces the possibility of false readings due to ghost voltages and improves accuracy when testing to determine absence or presence of voltage.

Safety consideration. Many people do not give much thought to safety when buying a DMM. However, there are large differences in how good DMMs are protected against common electrical hazards. It is very important to consider the environment and applications you will use in order to choose an appropriate device. With these safety criteria in mind, there would be no physical

damage to the device or life-threatening contact to electric energy while you hold it in your hand.

Voltage rating. It is important to identify the maximum voltage at which the circuit is designed to work. Choose a DMM rated to measure voltage that is expected to be present on the circuit.

Transient voltage rating. Such voltage comes from two main sources: natural causes such as lightning outside the building or by switching operations on the power-distribution system. Switching events in power distribution include switching of transformer taps, motors, inductances, sudden variation of load or disconnection of circuit breakers. Amplitudes of these voltages vary from a few hundred volts peak to about 6000 volts peak. These randomly-occurring high-voltage spikes tend to last from 50 microseconds to 200 microseconds.

Energy capacity. To protect yourself you should also know the energy capacity of the circuit before you start taking measurements. Circuits with higher energy capacity can deliver more current and energy into faults than low-energy circuits.

There are four categories (Table I) defined by International Electrotechnical Commission (IEC) that all manufacturers of DMMs are required to mark their products with. These are CAT I, CAT II, CAT III and CAT IV. With these markings one can easily identify what the maximum transient voltage is that the meter can safely withstand.

Safety indicators. Responsible manufacturers of DMMs obtain safety certifications from independent third-party testing agencies. These marks can usually be found at the back of the devices.

DMM probes should also be marked with a logo of a third-party safety-testing agency. You can check out some such marks in Fig. 2.

Other important features. Large displays, clear backlight to read comfortably in dark areas, ergonomic